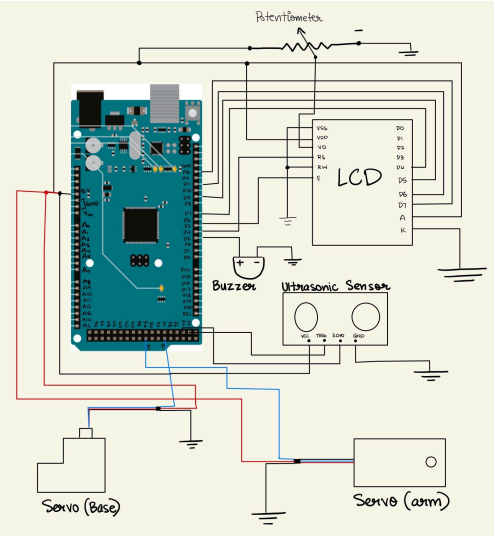
Rotating, Distance-sensing Turret

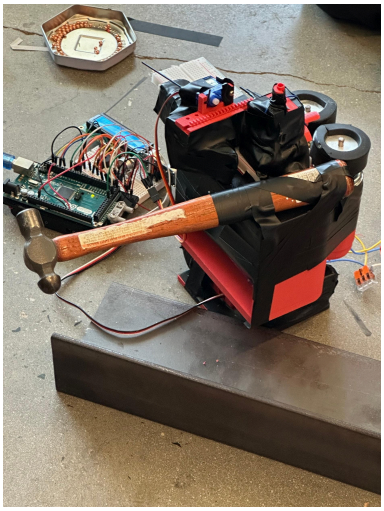
Members: Joshua Cherian, Aryan Vadrevu, Aagam Mehta, Cecilia O'Connell, Miranda Byszynski

Link to video: https://youtu.be/3SWY-vlytIE

For our project, our main idea was to initially create an RC car that had a turret on the top of it for shooting targets. However, after reassessing our limits, we decided to focus on the turret on top as our project.

Our main goals/specifications for this project were to create a turret that can: shoot BB pellets; rotate; detect when an object in front of it is within 30 centimeters, and if so, shoot a BB pellet at the object; display how many degrees it rotates on an LCD; and play an auditory signal when it is about to shoot.





All six mechatronic subsystems were employed in this project and formed an integral part of helping the overall success of the machine.

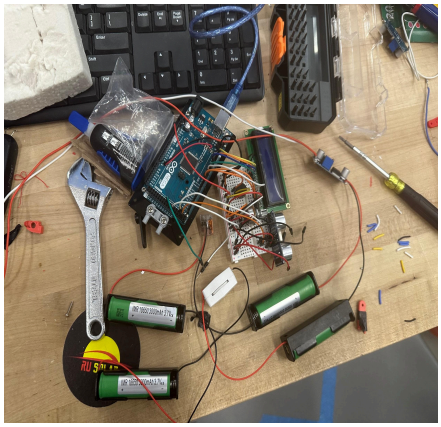
The first subsystem, an output display, was satisfied by our LCD. The LCD displayed the rotation in degrees of the turret upon the central axis, as designated by user input through the Serial Monitor on the Arduino application.

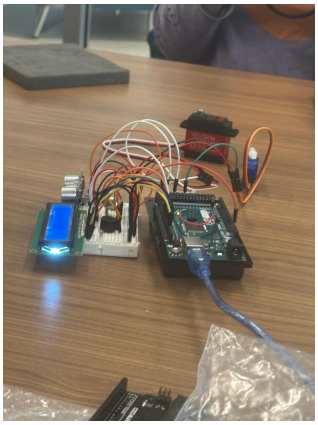
The second subsystem, an audio output device, was satisfied by our active buzzer. When the ultrasonic sensor sensed an object, a signal was sent so the buzzer would make a quick buzz while the turret would shoot at a target.

The third subsystem, a manual user input, was satisfied by our keyboard. Thanks to our Serial.available() coding, we were able to incorporate a user input. When

a user entered a number in the serial monitor, the turret would turn by said number of degrees.

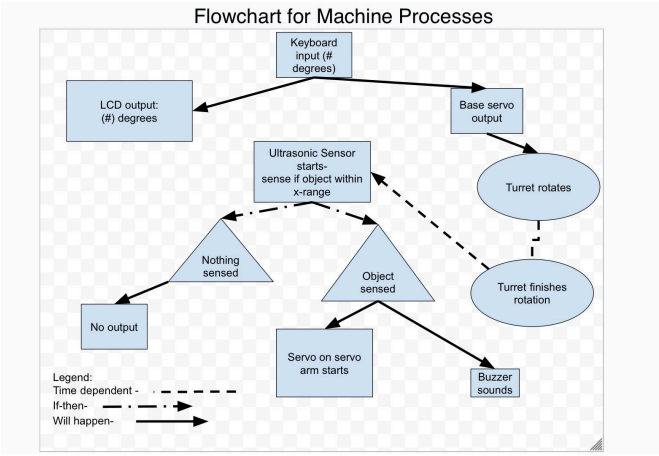
The fourth subsystem, an automatic sensor, was satisfied by our ultrasonic sensor. The sensor would sense how far any object in front of it was, information determining if the buzzer and shooter should turn on.





The fifth subsystem (actuators, mechanisms, and hardware) was satisfied by our servo motors and DC motors. 2 continuously running DC motors provided our BB projectiles with thrust to achieve quick forward motion. One servo was placed at the base of the turret to turn the entire launcher to the proper direction designated by user input. The second servo was used as an arm to push BBs towards the spinning DC motors; this servo did this through a linkage that converted its rotational motion to linear and pushed the projectiles.

The sixth subsystem, Logic Processing and Control, was satisfied by the Arduino coding used to direct everything. The main coding is described in the flowchart. The coding resets the arm and base servos and waits for input from the user. When the user inputs a number, the base servo turns by that much. After the base servo turns, the sensor senses how far the nearest object is. If the object is within a specified range of distance away, then the arm servo pushes the projectile forward and resets.



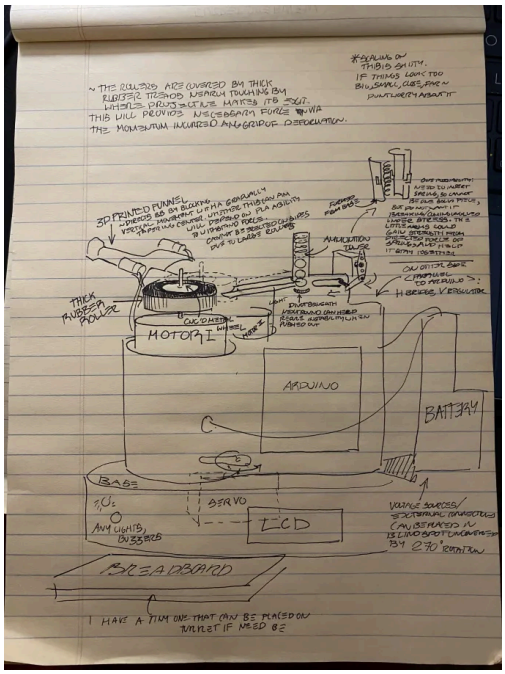

```
void loop()
{
while (Serial.available()== 0) {
}
userInput = Serial.parseInt();

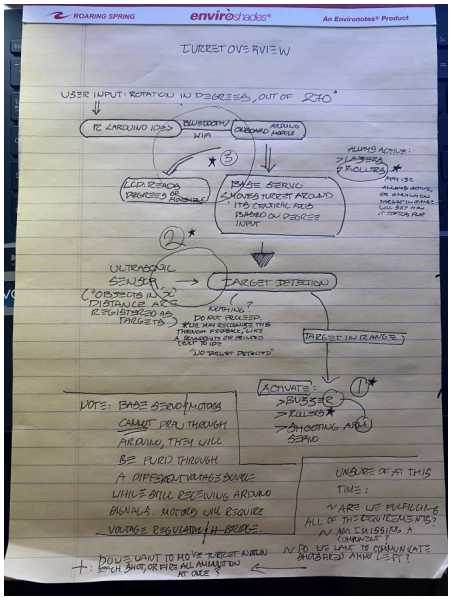
  if(userInput !=0)
  {
  Serial.println(userInput);
  delay(1000);
  baseServo.write(userInput);                 // sets the servo position according to the value
  delay(15);                                // waits for the servo to get there
  lcd.setCursor(0,0);
  lcd.print("Direct: ");
  lcd.setCursor(0,2);
  lcd.print(userInput);
  lcd.print(" deg" );
  }
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);


duration = pulseIn(echoPin, HIGH); //measures time sound wave took to travel
distance = (duration*.0343)/2; //converts time into centimeters
Serial.println(distance);

 unsigned char i;
 //output a frequency
 if (distance <= 30 )  //senses when object is within 30 cm to set off buzzer
 {
   for(i=0;i<100;i++)
   {
    digitalWrite(buzzer,HIGH);
    delay(1);//wait for 1ms
    digitalWrite(buzzer,LOW);
```

We used an Arduino Mega for the microcontroller to use most of our components. Excluding the rollers, the Arduino Mega provided power to and decided when and how each part would function. The ultrasonic sensor was powered and activated by the Arduino and then would send its results back to the microcontroller, so that the Arduino Mega could determine whether to activate the shooting mechanism and the buzzer. We had one servo (the base servo) that moved the entire shooting mechanism, and the degrees with which it rotated would be determined by the number a user typed into a computer, which would then be transmitted through the Arduino to become a motion in the base servo. The rollers were not turned on by the Arduino and were turned on separately. This was not an issue because, although the rollers were what gave the pellet enough forward motion to shoot outwards, the rollers did not start the shooting mechanism. Another servo (the arm servo) began the shooting mechanism. When activated, the arm servo pushed a pellet forward so it could roll towards the rollers, where it would be shot out. Then, the arm servo would return to its usual position.





This project was very challenging. However, in the end, we had a lot of success. We were able to get separate instances of the turret working when it needed to shoot and not shoot. The code worked very well. However, by the end of the project, the machine started having unknown problems with the arm servo. The coding was correct, but we needed to unplug and replug it. Next time, we should choose an easier project in terms of the six subsystems and try dividing our time better throughout the project.

In summary, this project delivered a functional and nearly automatic shooting apparatus. The idea was originally inspired by our interest in military applications, though it was not the main reason for our choosing this product. It consists of an LCD display, an automatic sensor, a buzzer, and a custom-built turret. This design was chosen for maximum efficiency and accuracy. Several improvements could have been made, however, such as optimizing sensor calibration, achieving full automation, refining control algorithms, and making the design more adaptable to various obstacles. Overall, this project has equipped us with various technical skills that can be used in a practical workspace, and that will be beneficial as we progress in our careers. This project fulfilled many requirements, such as logic and control, manual user inputs, etc, allowing meaningful hands-on experience and bolstering our problem-solving capabilities while simultaneously reinforcing our understanding of mechatronics systems as a whole.